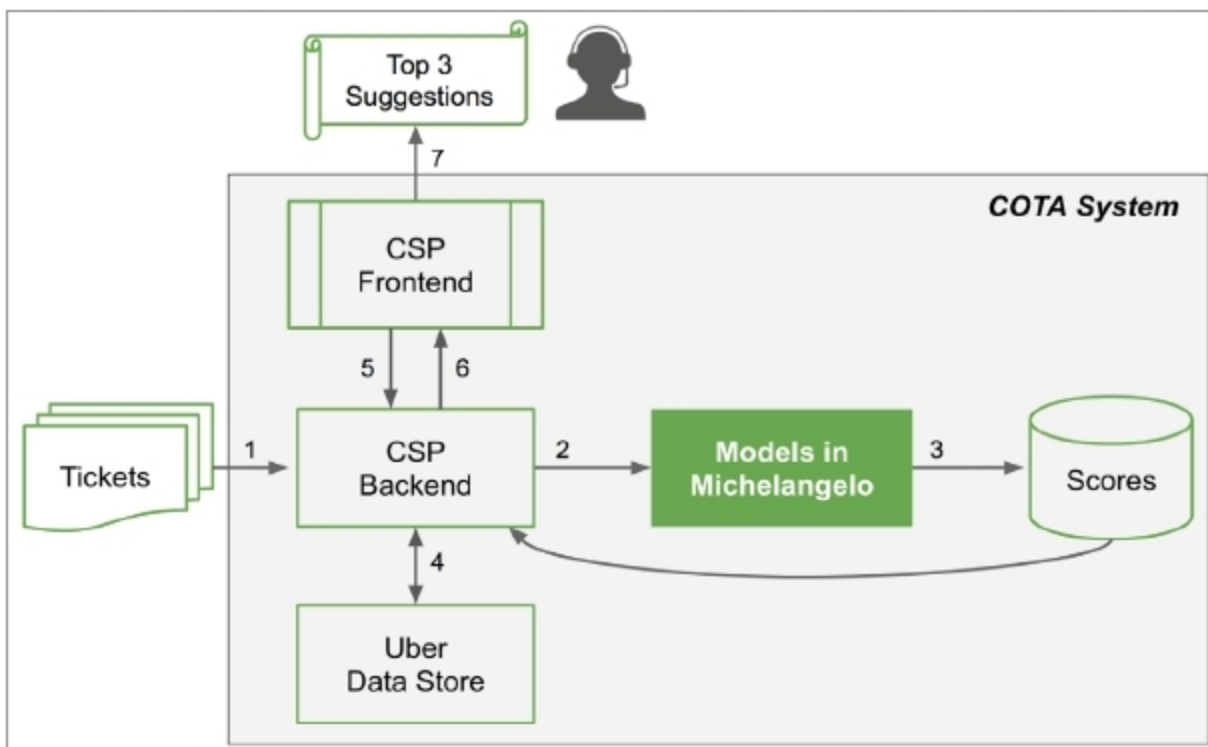




AI-based COTA by Uber

language understanding (NLU) based voice channel for communication works by converting audio to text format. For example, Dialogflow CX, launched by Google, is used for building text and voice-based AI agents for mobile apps, websites, and Google Assistant actions. A user can speak to the virtual agent naturally, and it will process the information and respond appropriately. Another example is the AI-powered voicebot TIA launched by Tata Capital on WhatsApp.

AI chatbot. The AI-based human-like bot that pops up on the website to start an online chat is becoming a common feature on most business websites. The chatbot can be integrated on websites, messaging applications, mobile apps, and kiosks. These bots play a vital role in the simplification of the interaction of humans with computers and customer interactions with the business, providing engaged self-service.

As per some reports, sixty per cent of online customers do not like to wait for more than sixty seconds for a response to their query. The use of chatbots drastically reduces customers' wait times and quicker resolution of their queries, thus improving customer experiences. Chatbots with natural language processing or NLP capabilities are used to handle customer complaints through faster responses, thus elevating customer satisfaction.

For example, the AI-based COTA (customer obsessed ticket assistant) system built by Uber helps customer

support representatives improve their speed and accuracy, resulting in improved customer experience. It has been found that with the AI system, ticket routing efficiency is increased by ten per cent.

Some big companies such as Google, Microsoft, Facebook, and others are all actively engaged in the race to build chatbots that can respond to customer queries.

Customer identification. These include biometrics, face, and voice recognition. Biometrics refers to body measurements, authentication, identification, and access control. Facial recognition identifies and verifies an individual by comparing facial features from a digital image or video to a database. Voice recognition biometrics digitises words and encodes them with data such as pitch, cadence, and tone, and then forms a unique voiceprint related to an individual.

AI-based biometrics helps customer service agents to recognise customers and greet them in a personal manner. The customer support team can use biometrics to quickly authenticate customers while minimising the risk of fraud.

For example, tools like ScopeAI work with customer support teams to analyse the voice of customer (VOC) data across multiple channels.

Predictive personalisation. Most companies now have access to huge amounts of data about their customers. When a customer clicks, views, and purchases a product from the

website, these data are translated into predictions that deliver value-added personalisation and recommendations to targeted consumers. That is, AI-based predictive technology can identify patterns that indicate a customer's intent based on Web activity or text, and route the call or chat to the appropriate agent.

AI has been used in the e-commerce industry to improve product recommendations based on personal customer needs, buying behaviour, and user profile. For example, Netflix's AI-powered customer service algorithm uses data such as demographics, viewing history, and personal preferences to predict what the user would like to watch next.

Predictive maintenance. This technology is used to predict technical and maintenance issues before major problems are developed. Some examples of AI predictive maintenance in use include United Parcel Service and Cisco Systems. It is reported that United Parcel Service has saved millions of dollars by implementing an AI predictive maintenance solution that reduces delivery truck breakdowns. In Cisco, AI predictive maintenance is used to optimise network performance and troubleshoot issues faster.

Business processes beyond customer support services suitable for RPA

So, are these business processes just specific to customer service? Absolutely not. Manish says, "Critical business processes such as contracting, compliance, customer service, salary payments are all tasks that are being automated. However, business processes such as finance and accounting—accounts payable, claims processing, data entry and reconciliation, document processing, and warehouse management—are a few areas that have witnessed the maximum deployment." A few examples of RPA usage in different fields beyond customer services are discussed next.

Banking and finance. It is one of the biggest sectors for RPA as myriads of outdated processes and systems with varying levels of complexity can be automated. It becomes simple to integrate data from different channels consisting of documents in different